

- 2 Artificial insemination (AI) is one method used in assisted reproduction programmes for large mammals.

For many years, horse breeders have collected semen samples from male horses and used these to inseminate female horses. Success rates have been good. However, due to the sample containing live sperm cells, the process needed to be carried out quickly.

New technologies exist to allow horse semen to be frozen in small plastic straws. Semen samples can now be stored for many years and transported all over the world.

- Each 0.5 cm^3 straw can hold 7.5×10^7 sperm cells.
- A typical sample of horse semen contains 7.5×10^9 sperm cells.

(a) (i) Calculate the volume of a typical horse semen sample.

..... cm^3 [1]

(ii) To inseminate one female horse, 5.0×10^8 sperm cells are needed.

Calculate the minimum number of straws needed to carry out this process.

minimum number = [1]

The semen from a horse is analysed by technicians to determine its quality before the horse is accepted onto the assisted reproduction programme.

Technicians use microscopy to look at the appearance and motility of the sperm cells and to estimate the sperm count.

Fig. 2.1 shows the surface view of a haemocytometer used to estimate the number of sperm cells in a sample.

Each number represents a sperm cell which was counted in the sample.

Each star (*) represents a sperm cell **not** counted in the sample.

The depth of the haemocytometer is 0.1 mm.

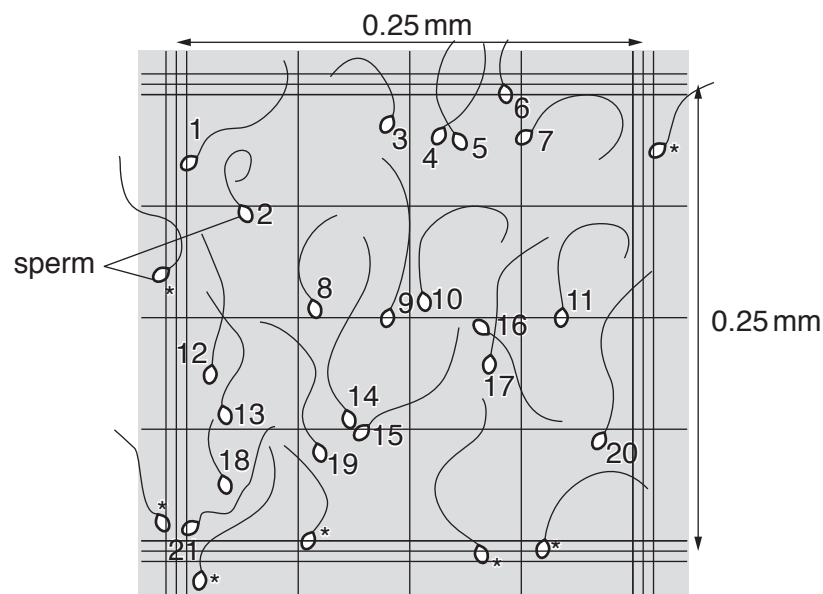


Fig. 2.1

- [4]

- The solution contains:

- Explain why it is necessary for the solution to contain these three substances.

sugar

.....

buffer

.....

antibiotics

.....

[3]

[Total: 9]